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C3 — Chemical Reactions

OCR GCSE Chemistry A (J248) — Gateway Science · Papers 1 & 3

Topic C3 is assessed on **Paper 1** (Foundation) and **Paper 3** (Higher), and the ideas here (formulae, equations, moles, acids, electrolysis) are assumed knowledge on Papers 2 and 4. Content marked **(HT)** is Higher Tier only — it matches the learning outcomes printed in bold in the specification.

C3.1 Introducing Chemical Reactions

> A **chemical reaction** makes **new substances**. The atoms in the reactants are **rearranged** — bonds are broken and new bonds are made — but no atoms are created or destroyed. A **chemical equation** shows this change in symbols.

Formulae of elements (C3.1a, C3.1c)

You will be given a Periodic Table in the exam. You should be able to use it to write symbols and formulae for at least the **first 20 elements**, the elements of **Groups 1, 7 and 0**, and any other common elements met in the course (e.g. Fe, Cu, Zn, Ag, Pb).

Type of element	How it is written	Examples
Metals (giant metallic lattice)	Just the symbol	Na, Mg, Fe, Cu, Zn
Giant covalent non-metals	Just the symbol	C (diamond, graphite), Si
Noble gases (Group 0, single atoms)	Just the symbol (monatomic)	He, Ne, Ar

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Diatomic molecules	Symbol with subscript 2	H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2

Exam Tip: Remember the seven diatomic elements: **H_2, N_2, O_2** and the halogens **F_2, Cl_2, Br_2, I_2**. Writing "Cl" for chlorine gas, or "O" for oxygen gas, in an equation loses the mark.

Formulae of simple covalent compounds (C3.1a)

Simple covalent compounds are made of small molecules. Their formula tells you exactly how many atoms of each element are in one molecule. Learn these common ones:

Name	Formula	Name	Formula
Water	H_2O	Carbon dioxide	CO_2
Ammonia	NH_3	Carbon monoxide	CO
Methane	CH_4	Sulfur dioxide	SO_2
Hydrogen chloride	HCl	Nitrogen dioxide	NO_2
Hydrogen peroxide	H_2O_2	Ethanol	C_2H_5OH

Prefixes in names tell you the numbers: **mon(o)-** = 1, **di-** = 2, **tri-** = 3, **tetra-** = 4. So dinitrogen tetroxide is N_2O_4.

Formulae of ionic compounds from ions (C3.1a, C3.1d)

An ionic compound has **no overall charge**, so the total positive charge must equal the total negative charge. Use the charges of the ions to work out how many of each you need.

Charge	Positive ions (cations)	Negative ions (anions)
1	H^+, Li^+, Na^+, K^+, Ag^+, NH_4^+ (ammonium), Cu^+	F^-, Cl^-, Br^-, I^-, OH^- (hydroxide), NO_3^- (nitrate), HCO_3^- (hydrogencarbonate)

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2	Mg^{2+}, Ca^{2+}, Ba^{2+}, Cu^{2+}, Zn^{2+}, Fe^{2+}, Pb^{2+}	O^{2-} (oxide), S^{2-} (sulfide), SO_4^{2-} (sulfate), CO_3^{2-} (carbonate)
3	Al^{3+}, Fe^{3+}	PO_4^{3-} (phosphate), N^{3-} (nitride)

Ion charges for simple ions follow from the group number: Group 1 → 1+, Group 2 → 2+, Group 3 → 3+, Group 6 → 2–, Group 7 → 1–. Transition metals can form more than one ion, which is why names include Roman numerals: iron(II) = Fe^{2+}, iron(III) = Fe^{3+}.

Worked Example: Write the formulae of (a) aluminium oxide, (b) calcium hydroxide, (c) ammonium sulfate.
(a) Al^{3+} and O^{2-}. Lowest common multiple of 3 and 2 is 6 → 2 Al^{3+} (+6) and 3 O^{2-} (–6) → **Al_2O_3**.
(b) Ca^{2+} and OH^- → one Ca^{2+} needs two OH^-. A compound ion that appears more than once goes in brackets → **$Ca(OH)_2$**.
(c) NH_4^+ and SO_4^{2-} → two NH_4^+ for one SO_4^{2-} → **$(NH_4)_2SO_4$**.

Exam Tip: Never change the formula of a compound ion: sulfate is always SO_4^{2-}. Use brackets when you need more than one of it, e.g. $Al_2(SO_4)_3$ — not $Al_2S_3O_{12}$.

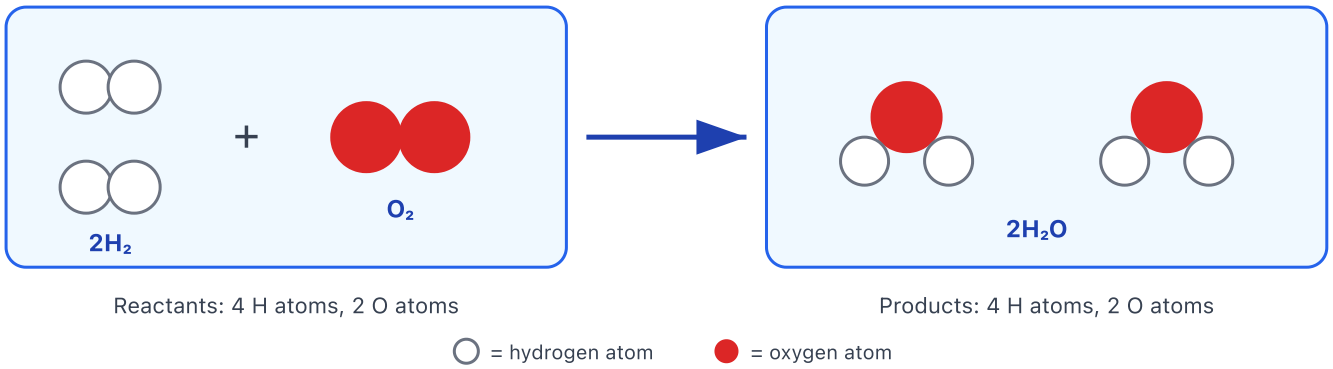
Writing balanced equations (C3.1b, C3.1c)

Because atoms are only rearranged, there must be the **same number of each type of atom on both sides**. This is the **principle of conservation of mass** written in symbols.

1. Write the word equation.
2. Write the correct formula for every reactant and product (do not change these later).
3. Count atoms of each element on each side.
4. Balance by putting **numbers in front of formulae** (coefficients) — never by changing subscripts.
5. Check every element again; leave elements that appear on their own (e.g. O_2) until last.

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Atoms are rearranged, not created or destroyed: $2H_2 + O_2 \rightarrow 2H_2O$



Worked Example: Balance: aluminium + oxygen → aluminium oxide.

Formulae: Al + O_2 → Al_2O_3. Oxygen: 2 on left, 3 on right → LCM 6 → $3O_2$ and $2Al_2O_3$. Now Al: 4 on right → 4Al.

$4Al + 3O_2 \rightarrow 2Al_2O_3$ (Check: Al 4 = 4, O 6 = 6 ✓)

State symbols (C3.1f)

Symbol	Meaning	Example
(s)	solid	Mg(s), $CaCO_3$(s), a precipitate such as AgCl(s)
(l)	liquid (pure liquid, not a solution)	H_2O(l), Br_2(l), molten NaCl(l)
(g)	gas	O_2(g), CO_2(g), H_2(g)
(aq)	aqueous — dissolved in water	NaCl(aq), HCl(aq), $CuSO_4$(aq)

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

Exam Tip: Water produced in a reaction is H_2O(l), not (aq). "Aqueous" means dissolved *in* water, so water itself can never be (aq). A molten ionic compound is (l).

Half equations **(HT)** (C3.1b)

A **half equation** shows what happens to **one species** in a reaction, with electrons (e^-) shown explicitly. They are used for redox reactions and electrolysis (see C3.3 and C3.4). Both **atoms** and **charge** must balance.

$$Na \rightarrow Na^+ + e^- \quad \text{(loss of electrons: oxidation)}$$
$$Cl_2 + 2e^- \rightarrow 2Cl^- \quad \text{(gain of electrons: reduction)}$$
$$Cu^{2+} + 2e^- \rightarrow Cu \qquad 2Br^- \rightarrow Br_2 + 2e^-$$

To check the charge: in $Cl_2 + 2e^- \rightarrow 2Cl^-$ the left side has total charge 2– (two electrons) and the right side has 2 × (1–) = 2–. ✓

Ionic equations (HT) (C3.1e)

In reactions in solution, some ions do not change — they are still dissolved, unchanged, at the end. These are **spectator ions**. An **ionic equation** leaves them out and shows only the particles that react.

1. Write the balanced full equation with state symbols.
2. Split every **(aq) ionic compound** into its ions. Do **not** split solids, liquids, gases or covalent molecules.
3. Cross out ions that appear unchanged on both sides.
4. Check atoms and charges balance.

Worked Example: Silver nitrate solution reacts with sodium chloride solution to form a white precipitate.
Full: $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$
Ions: $Ag^+(aq) + NO_3^-(aq) + Na^+(aq) + Cl^-(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3^-(aq)$
Spectators: Na^+ and NO_3^-. Ionic equation: **$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$**

Reaction type	Ionic equation
Any acid + alkali neutralisation	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

Metal displacement: zinc + copper(II) sulfate	$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$
Halogen displacement: chlorine + potassium bromide	$Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$
Acid + carbonate	$2H^+(aq) + CO_3^{2-}(aq) \rightarrow H_2O(l) + CO_2(g)$ (for a soluble carbonate)
Precipitation of copper(II) hydroxide	$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$

The mole and the Avogadro constant (HT) (C3.1g)

The **Avogadro constant** is the number of particles in one mole of a substance: **6.02×10^{23}** per mole.
One **mole** of a substance is the amount that contains **6.02×10^{23} particles** (atoms, molecules, ions or electrons — you must say which). Equivalently (classical definition), it is the amount containing as many particles as there are atoms in exactly 12 g of carbon-12. OCR accepts either definition.

The mass of one mole of a substance in grams is numerically equal to its **relative formula mass (M_r)** — or relative atomic mass (A_r) for an element made of atoms. So 1 mol of carbon atoms has mass 12 g, and 1 mol of water molecules has mass 18 g. Both contain 6.02×10^{23} particles.

$$\text{Number of particles} = \text{amount (mol)} \times 6.02 \times 10^{23}$$

$$\textbf{Mass of one particle (g)} = \mathbf{M_r \div 6.02 \times 10^{23}}$$

Worked Example: (a) Mass of one carbon atom: $12 \div (6.02 \times 10^{23}) = \mathbf{1.99 \times 10^{-23}\ g}$.
(b) Mass of one water molecule ($M_r = 18$): $18 \div (6.02 \times 10^{23}) = \mathbf{2.99 \times 10^{-23}\ g}$.
(c) Number of molecules in 0.25 mol of CO_2: $0.25 \times 6.02 \times 10^{23} = \mathbf{1.51 \times 10^{23}\ molecules}$. Each molecule has 3 atoms, so this is 4.52×10^{23} atoms.

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Exam Tip: Use your calculator's ×10ˣ (EXP) key for standard form. Give answers in standard form to 3 significant figures unless the question says otherwise — the data (6.02) is given to 3 s.f.

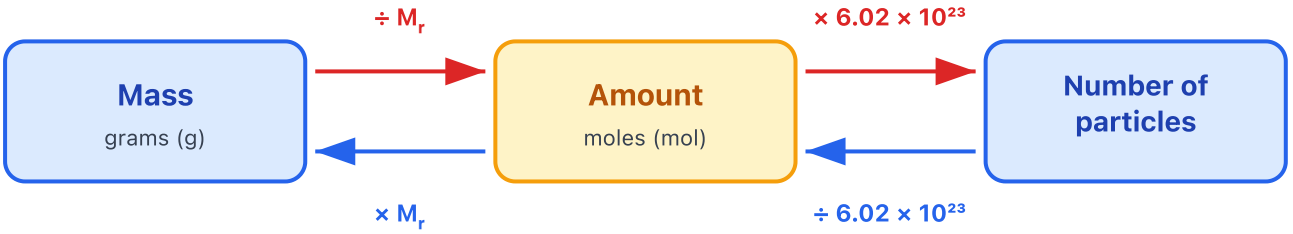
Mass and amount in moles (HT) (C3.1h)

Relative formula mass M_r is the sum of the relative atomic masses of all atoms in the formula. For example $M_r(CaCO_3) = 40 + 12 + (3 \times 16) = 100$, and $M_r(Ca(OH)_2) = 40 + 2 \times (16 + 1) = 74$.

$$\textbf{amount (mol)} = \textbf{mass (g)} \div \mathbf{M_r} \quad n = m \div M_r$$

$$\text{mass (g)} = \text{amount (mol)} \times M_r$$

Converting between mass, moles and number of particles (HT)



Always go through moles: you cannot jump straight from mass to number of particles.

Worked Example: (a) How many moles in 22 g of carbon dioxide? $M_r(CO_2) = 12 + 2 \times 16 = 44$. n = 22 ÷ 44 = **0.50 mol**.
(b) What is the mass of 0.25 mol of calcium carbonate? m = 0.25 × 100 = **25 g**.
(c) How many molecules in 9.0 g of water? n = 9.0 ÷ 18 = 0.50 mol → 0.50 × 6.02 × 10^{23} = **3.01 × 10^{23} molecules**.

Conservation of mass (C3.1i)

The **law of conservation of mass**: in a chemical reaction, **no atoms are lost or made**, so the **total mass of the products equals the total mass of the reactants**.

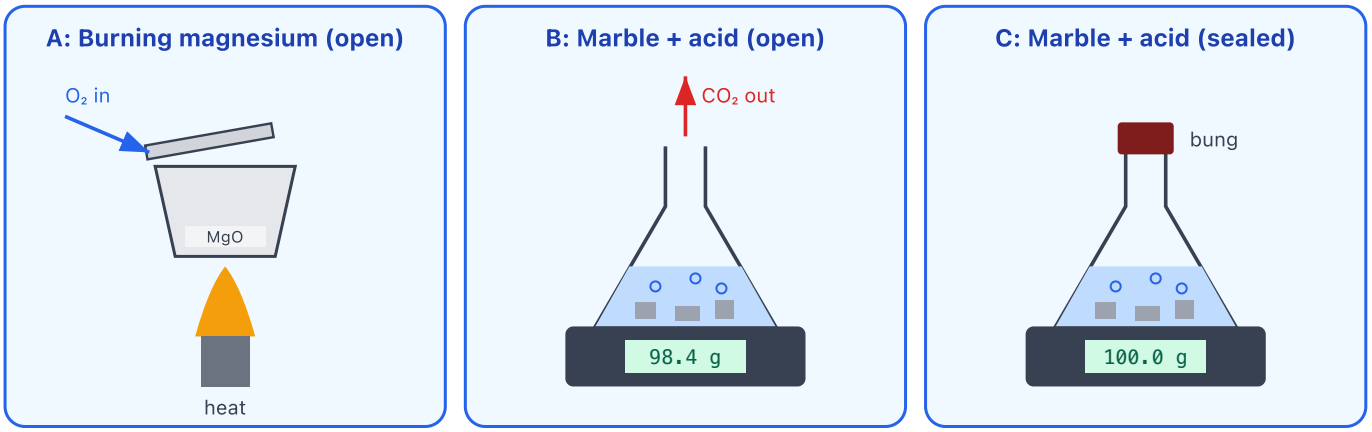
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Worked Example: 12.0 g of magnesium burns completely to form 20.0 g of magnesium oxide. What mass of oxygen reacted?
Mass of oxygen = 20.0 − 12.0 = **8.0 g**.

Observed mass changes in non-enclosed systems (C3.1j)

Sometimes the mass on a balance *appears* to change during a reaction. Mass is still conserved — the change happens because the system is **open** (non-enclosed), so **gas particles can enter or leave**.

Why the balance reading can change (open systems) or stay the same (closed)



Mass of solid INCREASES
O atoms from air join on

Reading DECREASES
CO_2 gas escapes into the air

Reading stays CONSTANT
no particles enter or leave

Observation (open container)	Example	Particle explanation
Mass increases	$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$; rusting of iron	A **gas from the air is a reactant**. Its atoms become part of the solid product, but the gas was never weighed at the start.
Mass decreases	$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$; acid + carbonate; acid + metal (H_2)	A **gas is a product**. Its particles spread out and escape from the container, so they are not weighed at the end.

No change	Precipitation: $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$	No gases are involved, so all particles stay in the container.

Exam Tip: If asked "why did the mass go down?", do not say "mass was lost" or "the substance disappeared". Say *"a gas (named) was produced and escaped from the open flask, so the mass of the gas was not recorded"*. In a **sealed** container the total mass is always constant.

Deducing stoichiometry from reacting masses **(HT)** (C3.1k)

The **stoichiometry** of a reaction is the ratio of the numbers of moles of each substance in the balanced equation. You can work it out from experimental masses:

1. Convert the mass of each reactant and product into moles ($n = m \div M_r$).
2. Divide every value by the smallest number of moles.
3. If necessary, multiply up to get whole numbers. These are the coefficients in the balanced equation.

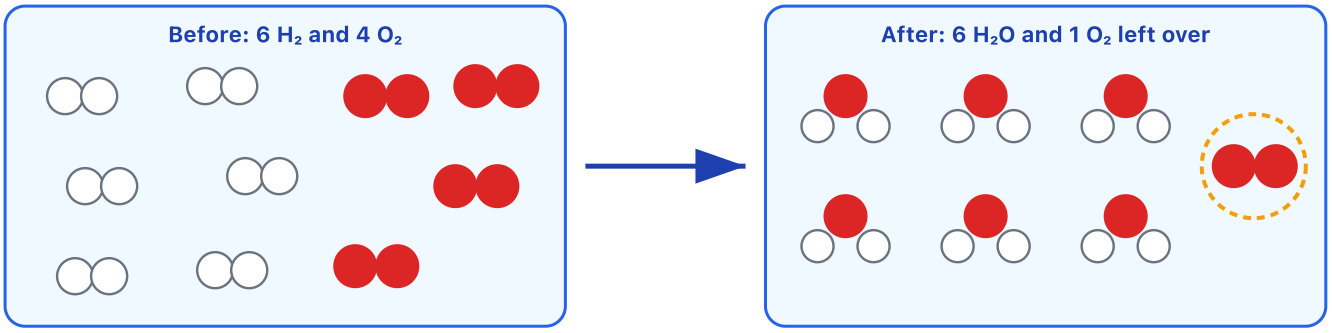
Worked Example: 5.60 g of iron reacts exactly with 10.65 g of chlorine, Cl_2, to form 16.25 g of iron chloride, $FeCl_3$. Deduce the balanced equation. (A_r: Fe = 56, Cl = 35.5)
$n(Fe) = 5.60 \div 56 = 0.100$ mol · $n(Cl_2) = 10.65 \div 71 = 0.150$ mol · $n(FeCl_3) = 16.25 \div 162.5 = 0.100$ mol
Ratio 0.100 : 0.150 : 0.100 = 1 : 1.5 : 1 = 2 : 3 : 2 → **$2Fe + 3Cl_2 \rightarrow 2FeCl_3$**
(Check conservation of mass: 5.60 + 10.65 = 16.25 g ✓)

Limiting reactants **(HT)** (C3.1k)

The **limiting reactant** is the reactant that is **completely used up**. It limits (decides) the amount of product formed. Any other reactant is **in excess** — some of it is left over at the end.

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Limiting reactant: 6 H_2 + 4 O_2 ($2H_2 + O_2 \rightarrow 2H_2O$)



6 H_2 need only 3 O_2

H_2 is LIMITING (all used up) · O_2 is in EXCESS

Adding more O_2 (the excess reactant) would make no more water; adding more H_2 would.

Worked Example: 4.0 g of hydrogen is mixed with 48.0 g of oxygen and ignited: $2H_2 + O_2 \rightarrow 2H_2O$. Which is limiting, and what mass of water forms?

$n(H_2) = 4.0 \div 2 = 2.0$ mol · $n(O_2) = 48.0 \div 32 = 1.5$ mol.

2.0 mol H_2 needs 1.0 mol O_2 (ratio 2 : 1). We have 1.5 mol, so O_2 is in excess (0.5 mol, 16 g, left over) and **H_2 is limiting**.

$n(H_2O) = n(H_2) = 2.0$ mol → mass = 2.0 × 18 = **36 g**.

Calculating reacting masses from equations **(HT)** (C3.1l)

mass of A → **moles of A** (÷ M_r) → **moles of B** (use the ratio in the equation) → mass of B (× M_r)

Worked Example: What mass of magnesium oxide forms when 6.0 g of magnesium burns?

$2Mg + O_2 \rightarrow 2MgO$ (A_r Mg = 24, O = 16)

$n(Mg) = 6.0 \div 24 = 0.25$ mol. Ratio Mg : MgO = 2 : 2 = 1 : 1, so $n(MgO) = 0.25$ mol.

$M_r(MgO) = 40$ → mass = 0.25 × 40 = **10 g**.

Worked Example: What mass of calcium oxide can be made by heating 1.0 tonne of calcium carbonate? $CaCO_3 \rightarrow CaO + CO_2$

Ratio 1 : 1 by moles. $M_r(CaCO_3) = 100$, $M_r(CaO) = 56$. So 100 g $CaCO_3$ → 56 g CaO, and 1.0 tonne → **0.56 tonnes** (560 kg). Using the same mass units on both sides saves converting to grams.

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C3.2 Energetics

> **Exothermic** reaction: energy is **transferred to the surroundings** — the **temperature of the surroundings rises**.
> **Endothermic** reaction: energy is **taken in from the surroundings** — the **temperature of the surroundings falls**.

Energy is never "used up" or "lost" — it is **transferred** between the chemicals (the system) and the surroundings (the solvent, container, air, your hand). The thermometer measures the surroundings.

Exothermic or endothermic? (C3.2a)

Exothermic (temperature rises)	Endothermic (temperature falls)
Combustion of fuels (e.g. methane, candle wax)	Thermal decomposition (e.g. $CaCO_3$ → CaO + CO_2)
Neutralisation (acid + alkali)	Citric acid + sodium hydrogencarbonate
Metals with acids; displacement reactions	Photosynthesis
Respiration; oxidation (e.g. hand warmers)	Electrolysis (needs an energy supply)

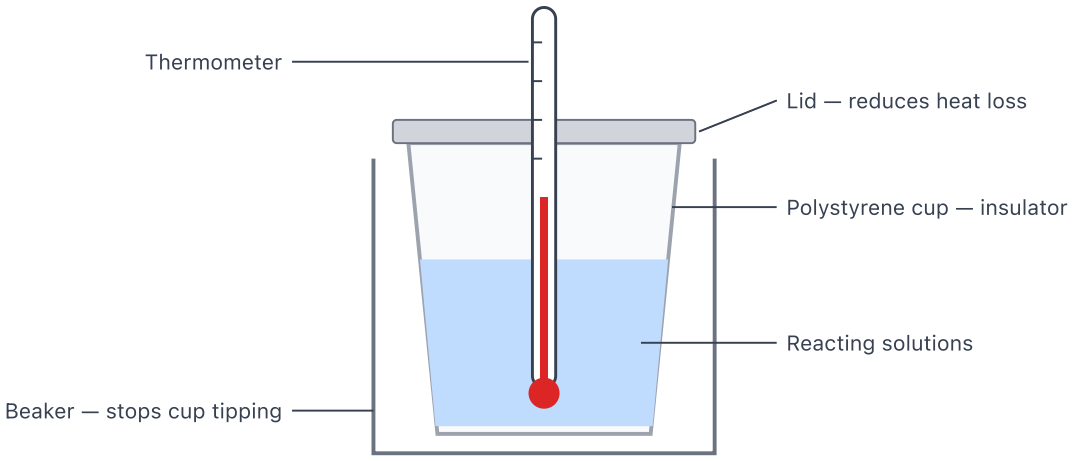
> **Exam Tip:** A burning candle is **exothermic** even though you need a match to light it — the match only supplies the *activation energy*. Overall, far more energy is given out than taken in.

PAG C8 link: measuring temperature changes

The temperature change of a reaction in solution is measured in an insulated **polystyrene cup** (OCR lists "Measuring the temperature change in reactions" under PAG C8).

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Measuring the temperature change of a reaction



Temperature change = highest (or lowest) temperature − starting temperature

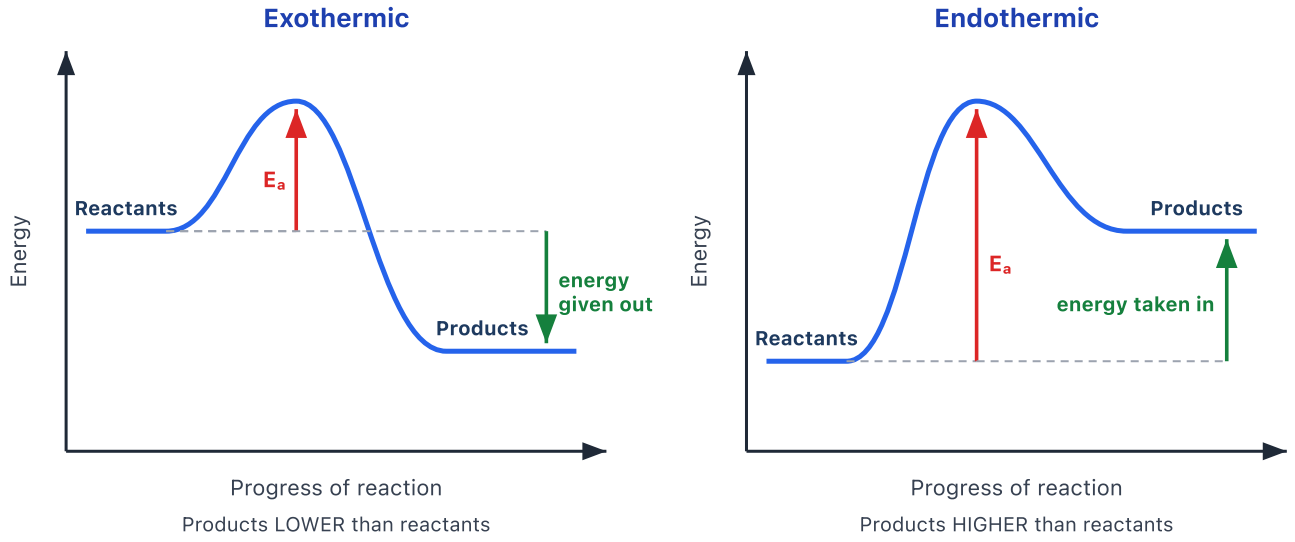
1. Measure a known volume (e.g. 25 cm^3) of one solution into the cup with a measuring cylinder; record the starting temperature.
2. Add a known volume/mass of the other reactant, put on the lid, stir.
3. Record the highest (exothermic) or lowest (endothermic) temperature reached.
4. Repeat and calculate a mean. Change one variable (e.g. mass of metal, concentration) to investigate its effect.

Reaction profiles and activation energy (C3.2b, C3.2c)

> **Activation energy (E_a)** is the **minimum energy needed for a reaction to occur** — the energy colliding particles must have to break bonds and start the reaction. On a reaction profile it is the height from the reactants to the top of the "hump".

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Reaction profiles



- Label: **reactants**, **products**, **activation energy** (arrow from reactants up to the peak) and **energy change** (arrow from reactants level to products level).
- Exothermic: products are lower — the overall energy change is negative ($\Delta H < 0$).
- Endothermic: products are higher — the overall energy change is positive ($\Delta H > 0$).
- A **catalyst** lowers the activation energy (a lower hump) but does not change the energy of reactants or products (see C5).

Bond breaking and bond making

During a reaction, bonds in the reactants are broken and new bonds in the products are made:

- **Breaking bonds is endothermic** — energy must be **taken in**.
- **Making bonds is exothermic** — energy is **given out**.

If more energy is released making the new bonds than is needed to break the old ones, the reaction is **exothermic** overall. If more energy is needed to break bonds than is released making them, it is **endothermic**.

> **Exam Tip:** A very common wrong answer is "energy is released when bonds break". It is the other way round: **breaking takes in, making gives out**.

Calculating energy changes from bond energies **(HT)** (C3.2d)

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Energy change = total energy needed to break bonds − total energy released making bonds

Negative answer → exothermic · Positive answer → endothermic (units: kJ/mol)

Bond	H–H	Cl–Cl	H–Cl	C–H	O=O	C=O	O–H
Bond energy (kJ/mol)	436	242	431	413	498	805	464

Typical values — always use the data given in the question.

Worked Example: $H_2 + Cl_2 \rightarrow 2HCl$. Calculate the energy change.
Bonds broken: 1 × H–H + 1 × Cl–Cl = 436 + 242 = 678 kJ/mol
Bonds made: 2 × H–Cl = 2 × 431 = 862 kJ/mol
Energy change = 678 − 862 = **−184 kJ/mol → exothermic**

Worked Example: Combustion of methane: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. Draw out the bonds first: CH_4 has 4 C–H; each O_2 has one O=O; CO_2 has 2 C=O; each H_2O has 2 O–H.
Broken: (4 × 413) + (2 × 498) = 1652 + 996 = 2648 kJ/mol
Made: (2 × 805) + (4 × 464) = 1610 + 1856 = 3466 kJ/mol
Energy change = 2648 − 3466 = **−818 kJ/mol → exothermic**

Exam Tip: Multiply each bond energy by the number of that bond in *one molecule* and then by the *coefficient* in the equation. Sketching displayed formulae (all bonds drawn) avoids missing bonds such as the double bonds in CO_2.

C3.3 Types of Chemical Reactions

Oxidation and reduction — oxygen (C3.3a)

Oxidation is the **gain of oxygen**. **Reduction** is the **loss of oxygen**. Reactions where both happen are called **redox** reactions.

An **oxidising agent** gives oxygen to another substance (and is itself reduced). A **reducing agent** removes oxygen from another substance (and is itself oxidised).

Reaction	Oxidised	Reduced	Reducing agent	Oxidising agent
$CuO + H_2 \rightarrow Cu + H_2O$	H_2 (gains O)	CuO (loses O)	H_2	CuO
$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$	CO	Fe_2O_3	CO	Fe_2O_3
$2CuO + C \rightarrow 2Cu + CO_2$	C	CuO	C	CuO
$2Mg + O_2 \rightarrow 2MgO$	Mg	O_2	Mg	O_2

Oxidation and reduction — electrons **(HT)** (C3.3b)

OIL RIG: **O**xidation **I**s **L**oss of electrons; **R**eduction **I**s **G**ain of electrons.
The **reducing agent** donates electrons (it is oxidised); the **oxidising agent** accepts electrons (it is reduced).

Worked Example: In $Cl_2 + 2KI \rightarrow 2KCl + I_2$, identify what is oxidised and what is reduced. Ionic: $Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$. Half equations: $Cl_2 + 2e^- \rightarrow 2Cl^-$ (gain → **chlorine reduced**, oxidising agent); $2I^- \rightarrow I_2 + 2e^-$ (loss → **iodide ions oxidised**, reducing agent).

Acids and alkalis (C3.3c)

Acids form **hydrogen ions, H^+(aq)**, when they dissolve in water.
Alkalis are soluble bases; their solutions contain **hydroxide ions, OH^-(aq)**.
A **base** is any substance that neutralises an acid (metal oxides, metal hydroxides, carbonates, ammonia). An alkali is a base that dissolves in water.

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Common acids: hydrochloric HCl (→ $H^+ + Cl^-$, makes chlorides), sulfuric H_2SO_4 (→ $2H^+ + SO_4^{2-}$, sulfates), nitric HNO_3 (→ $H^+ + NO_3^-$, nitrates), ethanoic CH_3COOH (partially → $H^+ + CH_3COO^-$, ethanoates).

Common alkalis: sodium hydroxide NaOH ($Na^+ + OH^-$), potassium hydroxide KOH, calcium hydroxide $Ca(OH)_2$ (limewater), and ammonia solution $NH_3(aq)$.

Neutralisation (C3.3d, C3.3e)

acid + alkali → salt + water acid + base (metal oxide / hydroxide) → salt + water

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

$$H_2SO_4(aq) + CuO(s) \rightarrow CuSO_4(aq) + H_2O(l)$$

Ionic equation for every aqueous neutralisation: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

The metal ion (or ammonium ion) in the salt comes from the base; the negative ion comes from the acid. All the other ions (e.g. Na^+ and Cl^-) are spectator ions, which is why **every** acid–alkali neutralisation can be generalised to hydrogen ions reacting with hydroxide ions to form water.

Acids with metals and carbonates (C3.3f)

acid + metal → salt + hydrogen $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$

acid + carbonate → salt + water + carbon dioxide $Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$

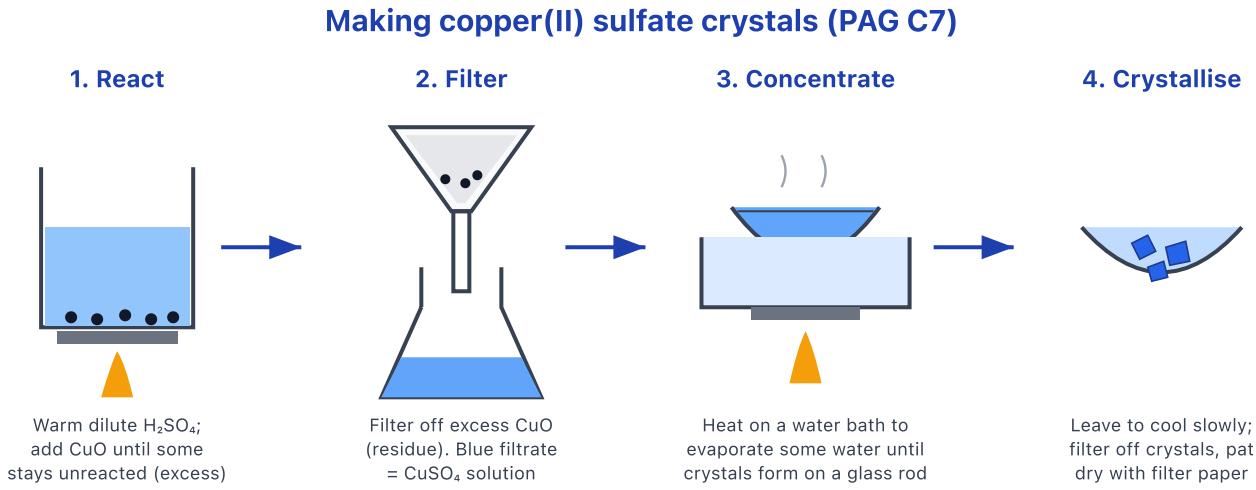
- Only metals **above hydrogen** in the reactivity series react with dilute acids (Mg, Zn, Fe do; Cu, Ag, Au do not). Very reactive metals (Na, K) react dangerously.
- Observations: **fizzing** (effervescence), solid disappearing, solution warming. Test: H_2 gives a squeaky pop; CO_2 turns limewater milky.

Worked Example: Predict the products and write a balanced equation for magnesium carbonate + nitric acid.
Salt: magnesium nitrate, $Mg(NO_3)_2$. **$MgCO_3 + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2O + CO_2$**

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PAG C7 link: production of a pure, dry salt

An **insoluble base** (metal oxide or carbonate) is added in **excess** to an acid. Example: copper(II) oxide + sulfuric acid → copper(II) sulfate + water.



Exam Tip: Why use **excess** base? To make sure **all the acid is used up**, so the salt is not contaminated with acid; the unreacted solid is easy to remove by filtration. Why not evaporate to dryness? Heating strongly can decompose hydrated salts and you get a powder, not crystals.

Dilute vs concentrated, strong vs weak **(HT)** (C3.3g)

Concentration = the **amount of acid (in moles, or mass) per unit volume of solution** (mol/dm³ or g/dm³). **Concentrated**: a large amount of acid in a given volume. **Dilute**: a small amount of acid in a given volume.
Strength = the **degree of ionisation**. A **strong acid** is **completely (fully) ionised** in water. A **weak acid** is only **partially ionised**.

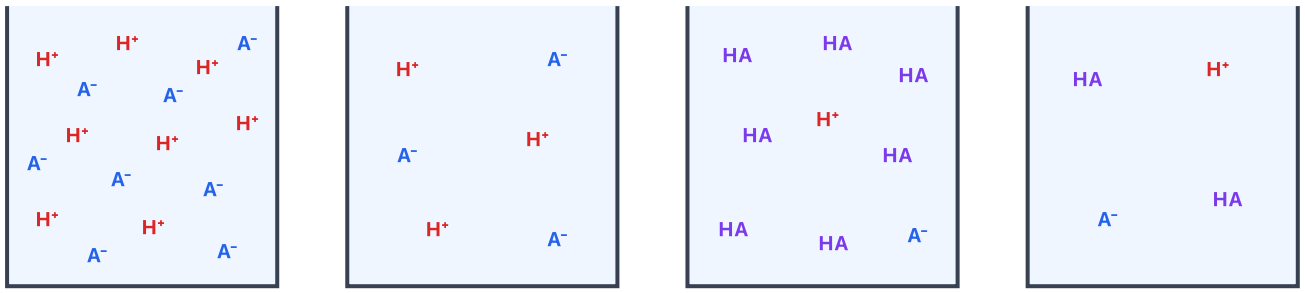
Strong: $HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$ (also HNO_3, H_2SO_4)

Weak: $CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$ (also citric acid, carbonic acid)

concentration (mol/dm³) = amount (mol) ÷ volume (dm³) · 1 dm³ = 1000 cm³

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Strength and concentration are different ideas (HT)



Concentrated, strong
many molecules, all ionised

Dilute, strong
few molecules, all ionised

Concentrated, weak
many molecules, few ionised

Dilute, weak
few molecules, few ionised

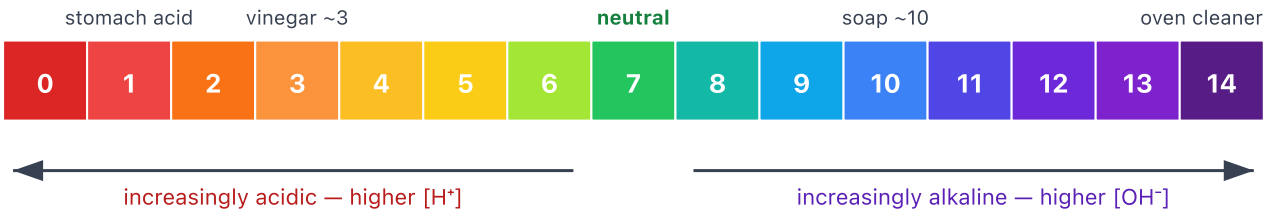
HA = un-ionised acid molecule · H^+ = hydrogen ion · A^- = negative ion from the acid

> **Exam Tip:** "Weak" does **not** mean "dilute". A concentrated solution of a weak acid can have a lower pH than a very dilute strong acid. At the **same concentration**, a strong acid has a higher H^+ concentration and so a **lower pH** than a weak acid, and reacts faster with metals and carbonates.

Measuring acidity: pH (C3.3h)

The **pH scale** (0–14) measures how acidic or alkaline a solution is. pH < 7 is acidic, pH 7 is **neutral**, pH > 7 is alkaline. The further from 7, the more acidic or alkaline.

The pH scale and universal indicator colours



Universal indicator: red/orange (acid) → green (neutral) → blue/purple (alkali). Exact shades vary by brand.

pH and hydrogen ion concentration **(HT)** (C3.3i, C3.3j)

- The **higher the concentration of H^+ ions**, the **lower the pH** (more acidic).
- In a **neutral** solution the concentrations of H^+ and OH^- are equal (pH 7).
- In an **alkaline** solution the H^+ concentration is very low (OH^- is in excess), so the pH is above 7.

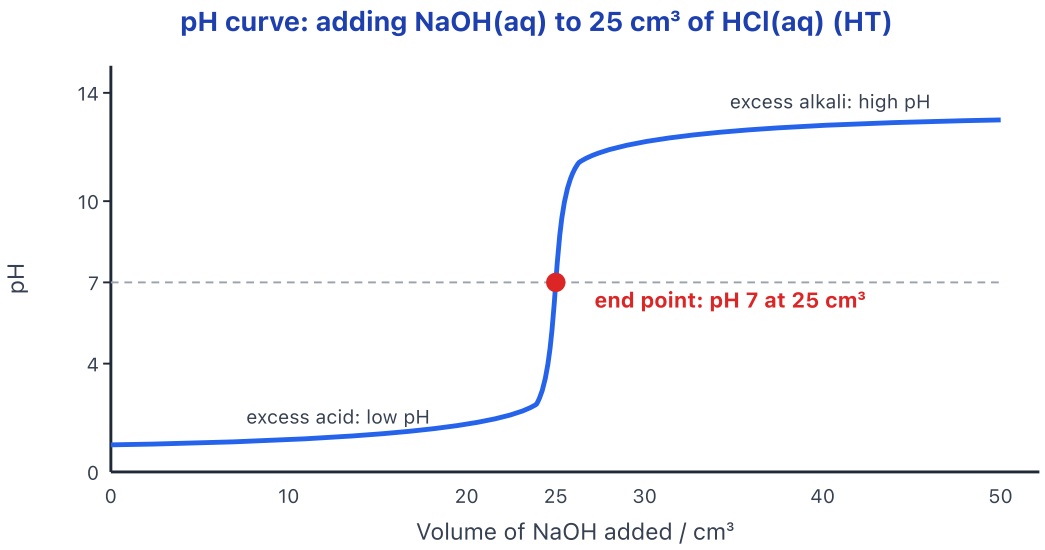
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- The scale is logarithmic: **each time the H^+ concentration increases by a factor of 10, the pH decreases by 1**.

pH	1	2	3	4	5
Relative [H^+]	10 000	1000	100	10	1

Worked Example: (a) Solution X has pH 2; solution Y has pH 5. How many times greater is [H^+] in X? Difference = 3 pH units → 10^3 = **1000 times**.
(b) A strong acid of pH 1 is diluted by a factor of 10 (10 cm^3 made up to 100 cm^3). [H^+] falls by a factor of 10, so the new pH = **2**. Diluting 100-fold would give pH 3.

(HT) pH titration curves. If a pH probe records pH while alkali is added from a burette to an acid, the pH rises slowly at first, then very steeply near the **end point** (when the acid is exactly neutralised), then levels off.



Measuring pH: techniques and apparatus (C3.3k)

Method	How	Advantages	Disadvantages
Universal indicator	Add a few drops (or dip paper/use a glass rod to spot);	Cheap, quick, simple	Only approximate (to the nearest whole number); colour judgement is

(solution or paper)	compare colour with a colour chart		subjective; hard to use with coloured solutions
pH meter / pH probe (with data logger)	Calibrate in buffer solutions; rinse probe with distilled water; place in solution and read display	Accurate (to 0.01); objective digital reading; can record continuously during a titration	More expensive; needs calibration

PAG C6 links: "Determining pH of unknown solutions", "Use of pH probes" and "Neutralisation reactions" — measure pH while adding alkali to acid; titrate a strong acid with a strong alkali using an indicator.

C3.4 Electrolysis

> **Electrolysis** is the **decomposition of an ionic compound, when molten or in aqueous solution, by passing an electric current through it**. The liquid or solution is the **electrolyte**.
> **Cathode** = negative electrode · **Anode** = positive electrode
> **Cations** = positive ions, attracted to the cathode · **Anions** = negative ions, attracted to the anode

Solid ionic compounds do **not** conduct: the ions are held in a fixed lattice and cannot move. When **molten or dissolved**, the ions are free to move and carry the charge. In the electrolyte the current is carried by **moving ions**, not electrons; electrons only flow in the external wires.

> **Exam Tip:** Memory aid: **PANIC** — **P**ositive **A**node, **N**egative **I**s **C**athode. Cations go to the cathode.

Products at the electrodes with inert electrodes (C3.4a)

- **Metals (or hydrogen) are formed at the cathode** — positive metal ions (or H^+) gain electrons.

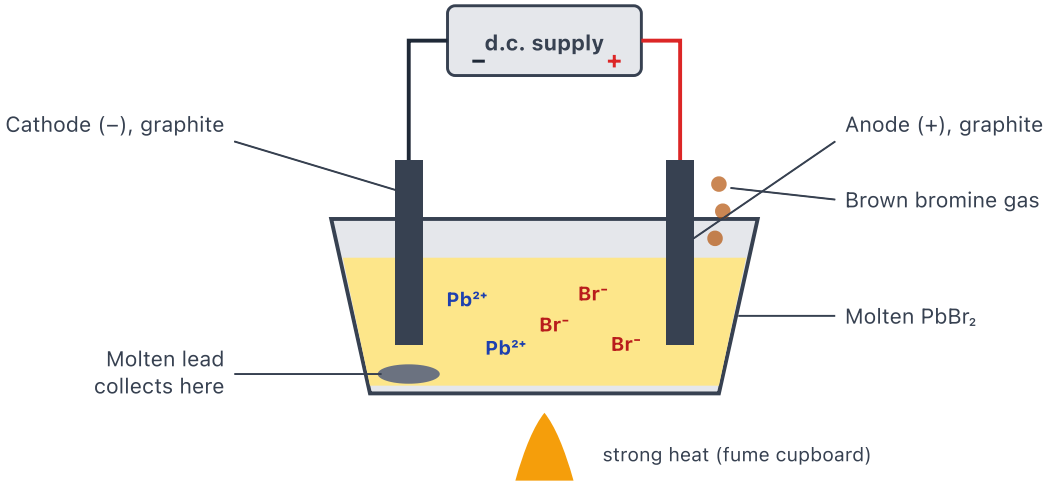
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- **Non-metals are formed at the anode** — negative non-metal ions lose electrons.
- **Inert electrodes** (graphite or platinum) do not react themselves.

Electrolysis of molten binary ionic compounds (C3.4b)

A **binary** compound contains just two elements. When molten, there are only two kinds of ion, so the products are simply the **metal at the cathode** and the **non-metal at the anode**.

Electrolysis of molten lead(II) bromide



Molten electrolyte	Ions present	Cathode (−) product	Anode (+) product
Sodium chloride, NaCl	Na^+, Cl^-	Sodium	Chlorine
Lead(II) bromide, $PbBr_2$	Pb^{2+}, Br^-	Lead	Bromine
Zinc chloride, $ZnCl_2$	Zn^{2+}, Cl^-	Zinc	Chlorine
Potassium iodide, KI	K^+, I^-	Potassium	Iodine
Aluminium oxide, Al_2O_3	Al^{3+}, O^{2-}	Aluminium	Oxygen

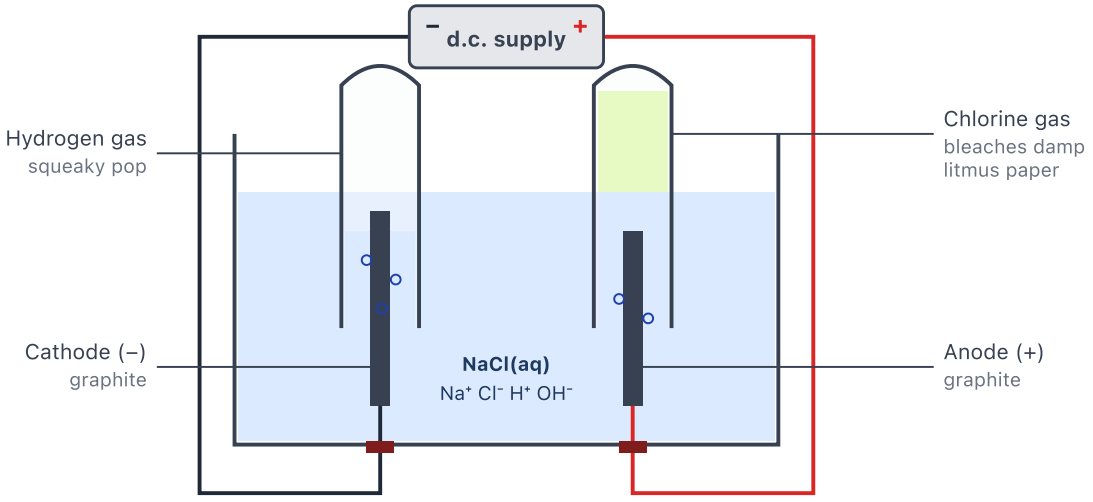
Electrolysis of aqueous solutions: competing reactions (C3.4c)

In an **aqueous solution** there are extra ions from water: a few water molecules ionise, $H_2O \rightleftharpoons H^+ + OH^-$. So **two cations compete** at the cathode and **two anions compete** at the anode. Only one of each is discharged.

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Electrode	Competing ions	Rule
Cathode (–)	Metal ion vs H^+	**Hydrogen** is produced unless the metal is **less reactive than hydrogen** (e.g. copper, silver, gold), in which case the **metal** is deposited.
Anode (+)	Non-metal ion vs OH^-	**Oxygen** is produced unless a **halide ion** (Cl^-, Br^-, I^-) is present, in which case the **halogen** is produced. (Sulfate and nitrate ions are never discharged.)

Electrolysis of sodium chloride solution, collecting the gases (PAG C2)



Less chlorine than hydrogen is collected at first, because chlorine is slightly soluble in water.

Aqueous electrolyte (inert electrodes)	Ions present	Cathode (–)	Anode (+)	Left in solution
Sodium chloride, NaCl(aq)	Na^+, H^+, Cl^-, OH^-	Hydrogen (Na is more reactive than H)	Chlorine (halide present)	Sodium hydroxide, NaOH
Copper(II) sulfate, $CuSO_4$(aq)	Cu^{2+}, H^+, SO_4^{2-}, OH^-	Copper (pink-brown coating; Cu less reactive than H)	Oxygen (no halide)	Sulfuric acid, H_2SO_4 (blue colour fades as Cu^{2+} is removed)

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Dilute sulfuric acid, $H_2SO_4(aq)$	H^+, SO_4^{2-}, OH^-	Hydrogen	Oxygen	Sulfuric acid (more concentrated)
Silver nitrate, $AgNO_3(aq)$	Ag^+, H^+, NO_3^-, OH^-	Silver	Oxygen	Nitric acid

PAG C2: electrolysis of aqueous solutions

Set up the cell above with graphite electrodes and a low-voltage d.c. supply (4–6 V); collect gases in inverted test tubes and test them (H_2: squeaky pop; O_2: relights a glowing splint; Cl_2: bleaches damp litmus). Observe any coating on the cathode. Chlorine is toxic — use a well-ventilated room, short run times and eye protection.

Equations at the electrodes (C3.4d)

At the **cathode**, positive ions **gain electrons** (reduction). At the **anode**, negative ions **lose electrons** (oxidation). Overall, the electrolysis of NaCl(aq) can be summarised as:

$$2NaCl(aq) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g) + Cl_2(g)$$

(HT) Half equations at the electrodes:

Electrode	Half equation	Type
Cathode: sodium from molten NaCl	$Na^+ + e^- \rightarrow Na$	Reduction (gain of e^-)
Cathode: lead from molten $PbBr_2$	$Pb^{2+} + 2e^- \rightarrow Pb$	
Cathode: hydrogen	$2H^+ + 2e^- \rightarrow H_2$	
Cathode: copper	$Cu^{2+} + 2e^- \rightarrow Cu$	
Anode: chlorine	$2Cl^- \rightarrow Cl_2 + 2e^-$	Oxidation (loss of e^-)
Anode: bromine	$2Br^- \rightarrow Br_2 + 2e^-$	
Anode: oxygen	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$	

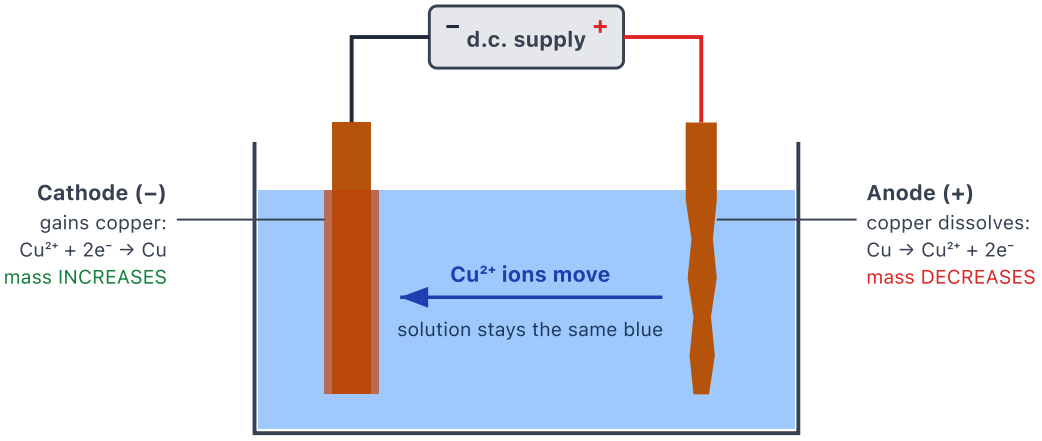
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Exam Tip: (HT) Check that the electrons balance: in $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$ the charge on the left is 4–, and on the right 0 + 0 + 4– = 4–. Electrons can also be written as a subtraction, e.g. $2Cl^- - 2e^- \rightarrow Cl_2$, which OCR accepts.

Inert and non-inert electrodes (C3.4e)

With **inert** electrodes (graphite, platinum) the electrodes just carry the current. With **non-inert** (active) electrodes, the electrode itself takes part. The key example is **copper(II) sulfate solution with copper electrodes**:

Electrolysis of $CuSO_4$(aq) with copper (non-inert) electrodes



Impurities from an impure copper anode fall to the bottom as "anode sludge" — this is how copper is purified.

Compare with **inert** electrodes in $CuSO_4$(aq): copper is still deposited at the cathode, but **oxygen** bubbles off at the anode, the blue colour **fades** and the solution becomes acidic. With copper electrodes there is no gas, the colour stays the same, and the cathode gain equals the anode loss. Copper electrodes are used to **purify copper**, and the same set-up (object to be coated as the cathode) is used for **electroplating**.

Practicals linked to C3 (PAGs)

PAG	Title (from the spec)	C3 activity	Key points

PAG C2	Electrolysis	Electrolysis of sodium chloride solution and of copper sulfate solution	Inert electrodes; collect and test gases; observe copper coating
PAG C6	Titration	Neutralisation reactions; determining pH of unknown solutions; use of pH probes	Indicator or pH probe; record pH/volume; end point at pH 7 for strong acid + strong alkali
PAG C7	Production of salts	Production of a pure, dry sample of a salt	Excess insoluble base, filter, concentrate on water bath, crystallise, dry
PAG C8	Measuring rates of reaction	Measuring the temperature change in reactions	Polystyrene cup + lid; record start and maximum/minimum temperature

Quick Summary — Key Equations for C3

Quantity	Equation	Units
Amount of substance (HT)	$n = m \div M_r$	mol
Number of particles (HT)	$N = n \times 6.02 \times 10^{23}$	—
Mass of one particle (HT)	$M_r \div 6.02 \times 10^{23}$	g
Conservation of mass	total mass of reactants = total mass of products	g
Concentration (HT)	amount (mol) ÷ volume (dm^3)	mol/dm^3
Energy change (HT)	bonds broken – bonds made	kJ/mol
Neutralisation (ionic)	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$	—
pH (HT)	$[H^+] \times 10 \rightarrow pH - 1$	—

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Specification checklist

Code	Learning outcome (summary)	Tier
C3.1a	Use symbols to write formulae of elements and simple covalent and ionic compounds	F + H
C3.1b	Use names, symbols and conservation of mass to write formulae and balanced equations (and half equations)	F + H (half equations HT)
C3.1c	Use a supplied Periodic Table to write formulae and balanced equations	F + H
C3.1d	Use formulae of common ions to deduce formulae of compounds	F + H
C3.1e	Construct balanced ionic equations	HT
C3.1f	State symbols (s), (l), (g), (aq)	F + H
C3.1g	Definitions of the Avogadro constant (6.02×10^{23}) and the mole; mass of one atom/molecule	HT
C3.1h	Relate mass of a substance to amount in moles and vice versa	HT
C3.1i	Recall and use the law of conservation of mass	F + H
C3.1j	Explain mass changes in non-enclosed systems using the particle model	F + H
C3.1k	Deduce stoichiometry from reacting masses; effect of a limiting reactant	HT
C3.1l	Use a balanced equation to calculate masses of reactants or products	HT
C3.2a	Distinguish exothermic and endothermic reactions by temperature change of surroundings	F + H

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C3.2b	Draw and label reaction profiles (activation energy, energy change, reactants, products)	F + H
C3.2c	Activation energy = energy needed for a reaction to occur	F + H
C3.2d	Calculate energy changes from bond breaking and bond making energies	HT
C3.3a	Oxidation/reduction as gain/loss of oxygen; oxidising and reducing agents	F + H
C3.3b	Oxidation/reduction as loss/gain of electrons	HT
C3.3c	Acids form H^+ in water; alkalis contain OH^-	F + H
C3.3d	Neutralisation: acid + alkali/base → salt + water	F + H
C3.3e	Aqueous neutralisation: $H^+ + OH^- \rightarrow H_2O$	F + H
C3.3f	Carbonates and some metals react with acids; balanced equations	F + H
C3.3g	Dilute/concentrated (amount) and weak/strong (degree of ionisation)	HT
C3.3h	Acidity and alkalinity are measured by pH	F + H
C3.3i	Neutrality, acidity and alkalinity in terms of H^+ concentration and pH; pH titration curves	HT
C3.3j	$[H^+]$ increases × 10 → pH decreases by 1	HT
C3.3k	Techniques and apparatus for measuring pH (universal indicator, pH meters)	F + H
C3.4a	Metals/hydrogen at cathode, non-metals at anode (inert electrodes); cations and anions	F + H
C3.4b	Predict products of electrolysis of molten binary ionic compounds	F + H

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C3.4c	Competing reactions in aqueous electrolysis (NaCl(aq), $CuSO_4$(aq))	F + H
C3.4d	Electrolysis in terms of ions and electrode reactions (half equations)	F + H (half equations HT)
C3.4e	Electrolysis with inert and non-inert electrodes	F + H

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